MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) FOURTH
SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

Course Code: **MG4DSCBOT201** Course Title: **Plant Anatomy and Reproductive Botany**

Time: 1.5 Hrs Maximum Marks: 50

Student should attempt at least one question from each course outcome to enhance their overall outcome attainability

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Part A: Short answers

Answer 10 questions
Each question carries 1 mark

	Question	Taxonomy	CO
1	Name a stain used in light microscopy.	[K]	1
2	Mention the role of a fixative in specimen preparation for microscopy.	[U]	1
3	What are plasmodesmata?	[K]	2
4	What are bordered pits?	[K]	2
5	Mention any two advantages of bioethanol over conventional fossil fuels.	[U]	2
6	Write any two conversion techniques used to convert cellulose into bioethanol.	[K]	2
7	Define dendrochronology.	[K]	3
8	What is heartwood?	[K]	4
9	What is meant by grain in wood?	[K]	5
10	Define polyembryony.	[K]	6
11	What is double fertilization?	[K]	6
12	Distinguish between cleistogamy and geitnogamy.	[U]	6

Part B: Short Essay Type Questions

Answer 6 questions
Each question carries 5 marks

13	What are secretory tissues? Explain any two with suitable examples.	[U]	1
14	Give a detailed account of the ultrastructure of the plant cell wall.	[U]	2
15	Discuss the scope of cellulose as a potential energy source for the future.	[U]	2
16	What is reaction wood? Compare tension wood and compression wood.	[U]	4
17	Describe the anatomical features used in the identification of commercial wood.	[U]	5
18	Explain in detail the applications of wood anatomy.	[U]	5
19	What is endosperm? Explain the different types of endosperm development.	[K]	6
20	Describe the internal structure of a dithecous anther.	[K]	6

Part C: Essay Type Questions

Answer any 1 question

Question carries 10 marks

21	Describe and illustrate the monosporic type of embryo sac development.	[U]	6
22	With a neat labelled diagram, explain the process of secondary thickening in a dicot stem.	[U]	3